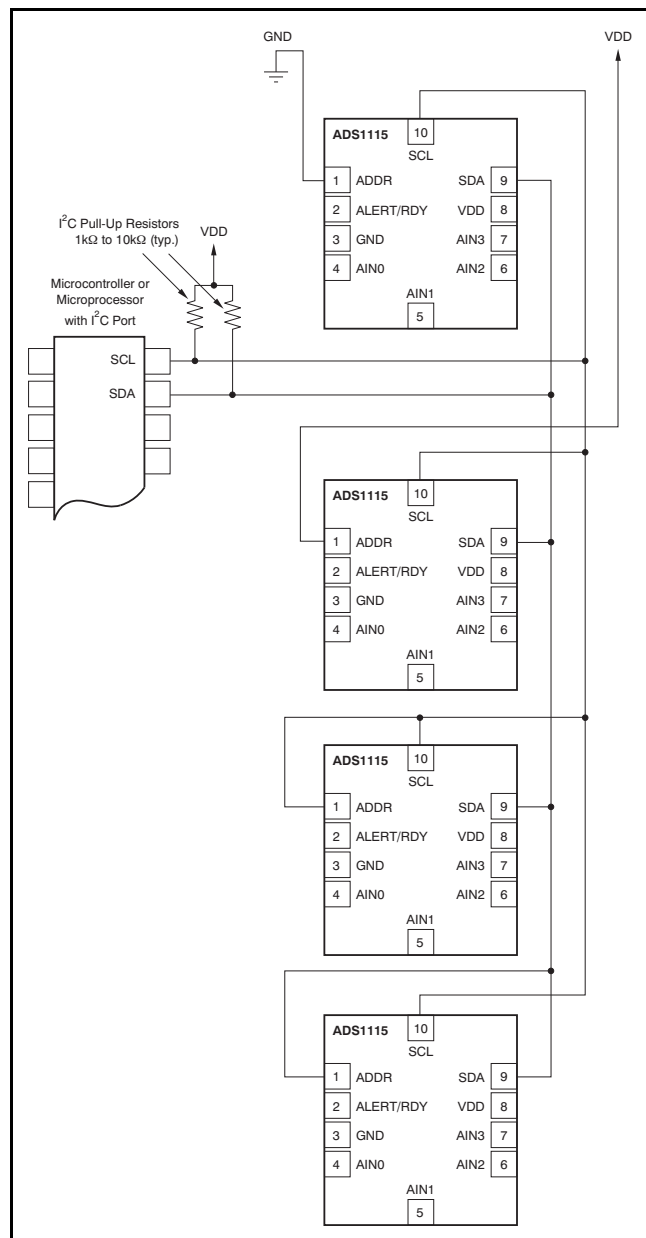
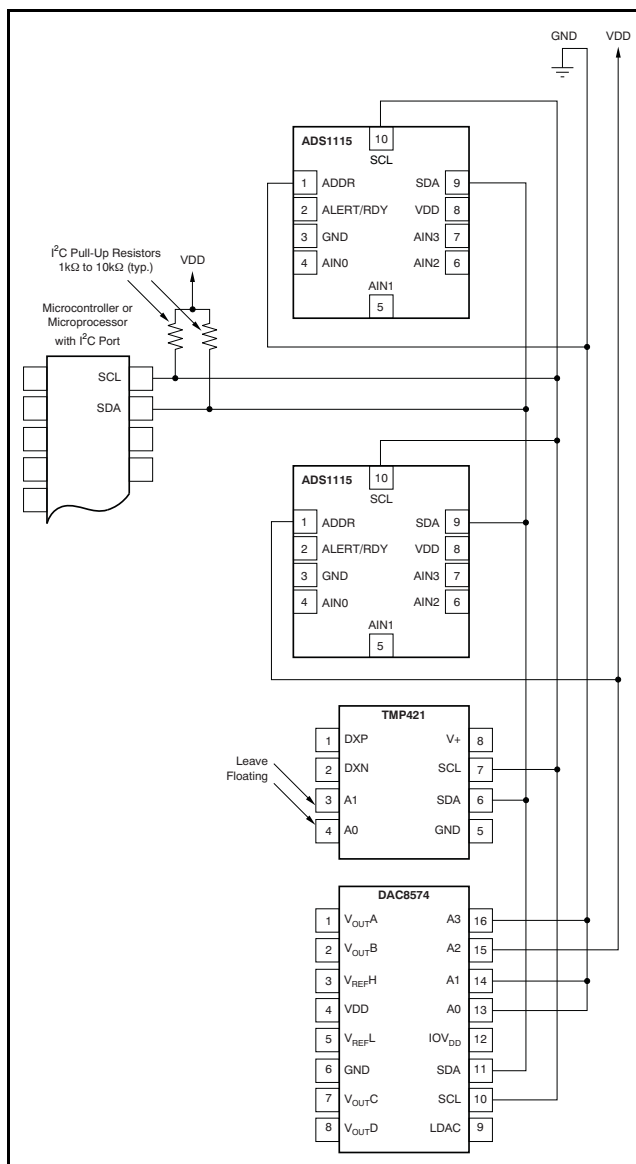




NOTE: ADS1113/4/5 power and input connections omitted for clarity. The ADDR pin selects the I²C address.

Figure 35. Connecting Multiple ADS1113/4/5s



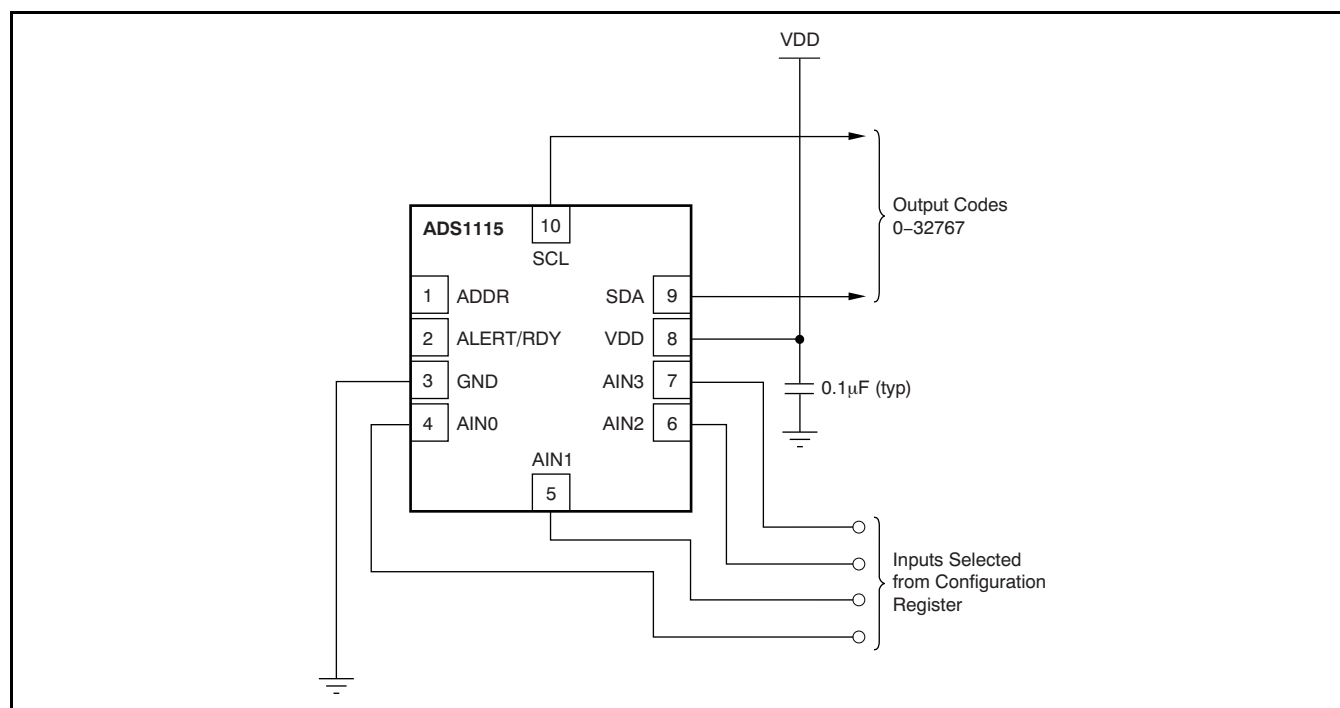
NOTE: ADS1113/4/5 power and input connections omitted for clarity. ADDR, A3, A2, A1, and A0 select the I²C addresses.

Figure 36. Connecting Multiple Device Types

SINGLE-ENDED INPUTS

Although the ADS1115 has two differential inputs, the device can easily measure four single-ended signals. [Figure 37](#) shows a single-ended connection scheme. The ADS1115 is configured for single-ended measurement by configuring the MUX to measure each channel with respect to ground. Data are then read out of one input based on the selection on the configuration register. The single-ended signal can range from 0V to supply. The ADS1115 loses no linearity anywhere within the input range. Negative voltages cannot be applied to this circuit because the ADS1115 can only accept positive voltages.

The ADS1115 input range is bipolar differential with respect to the reference. The single-ended circuit shown in [Figure 37](#) covers only half the ADS1115 input scale because it does not produce differentially negative inputs; therefore, one bit of resolution is lost.



NOTE: Digital and address pin connections omitted for clarity.

Figure 37. Measuring Single-Ended Inputs